CA Dissolution–Precipitation Model

All Figures – Task 1 (Base Case Explorations) and Task 2 (Weighted Walk Comparison)

Sneha Chakrabarti IISER Kolkata May–June 2026

Supervised by Prof. Vaughan Voller (University of Minnesota) and Prof. Piotr Szymczak (University of Warsaw)

Fixed base parameters: $N = 201 \times 201 \cdot N_{\text{part}} = 4000 \cdot$ `diss_rate` $= 0.05 \cdot C_s = 1.0 \cdot$ seed $= 42$
Colour: navy = dissolved (M reduced) · yellow = precipitated ($P = 1$) · teal = fresh mineral

Task 1 – Base Case Explorations `base_case_explorations.py`

Uniform walk: equal probability to each open downstream neighbour. Controlling group: $S = C_s/$`diss_rate` (steps per particle).

Exploration 1 – Symmetry

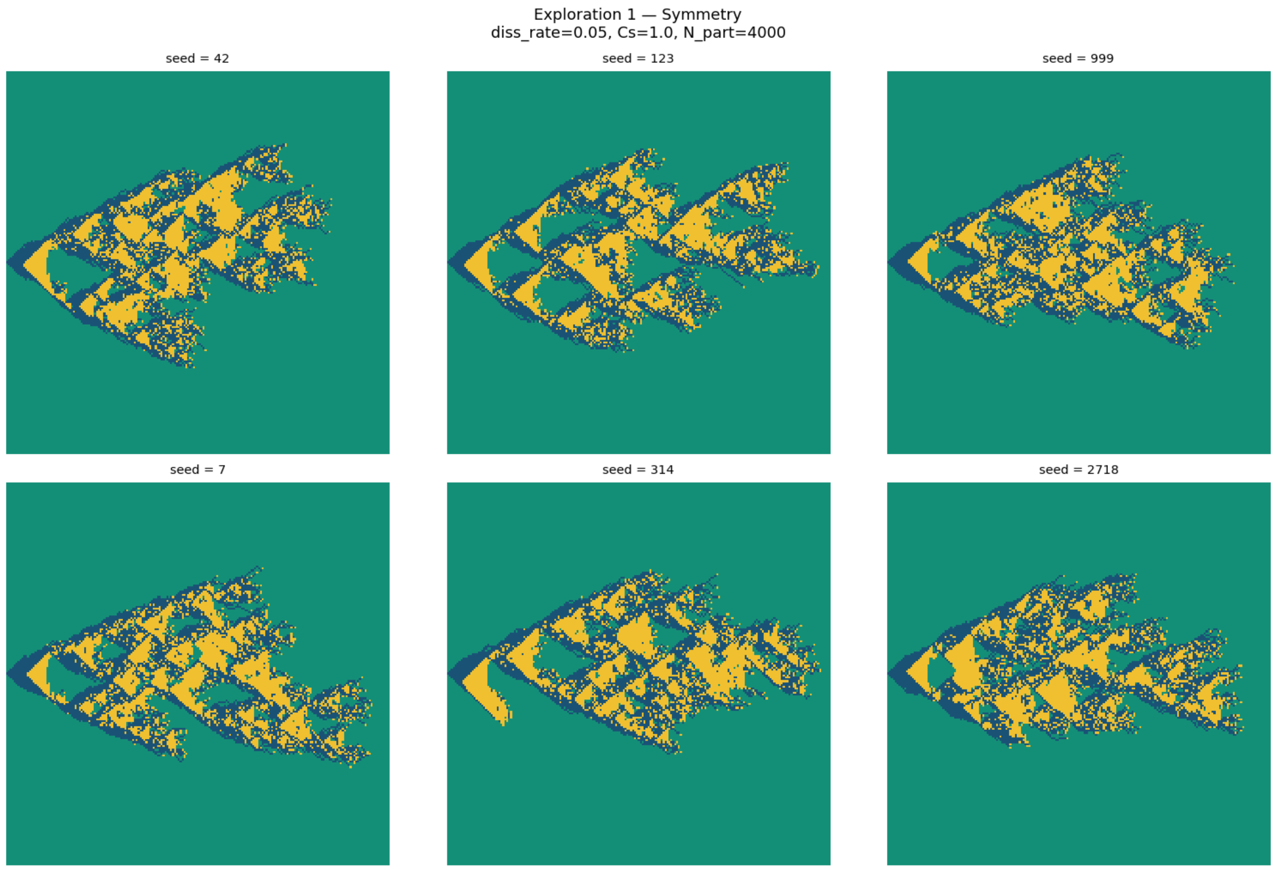

Figure 1: Six independent RNG seeds; `diss_rate`$= 0.05$, $C_s = 1$, $N_{\text{part}} = 4000$. Asymmetry about $y = 100$ changes with seed, confirming it is stochastic rather than structural. Larger N_{part} or ensemble averaging recovers the expected up–down symmetry.

Exploration 2 – Saturation Threshold C_s

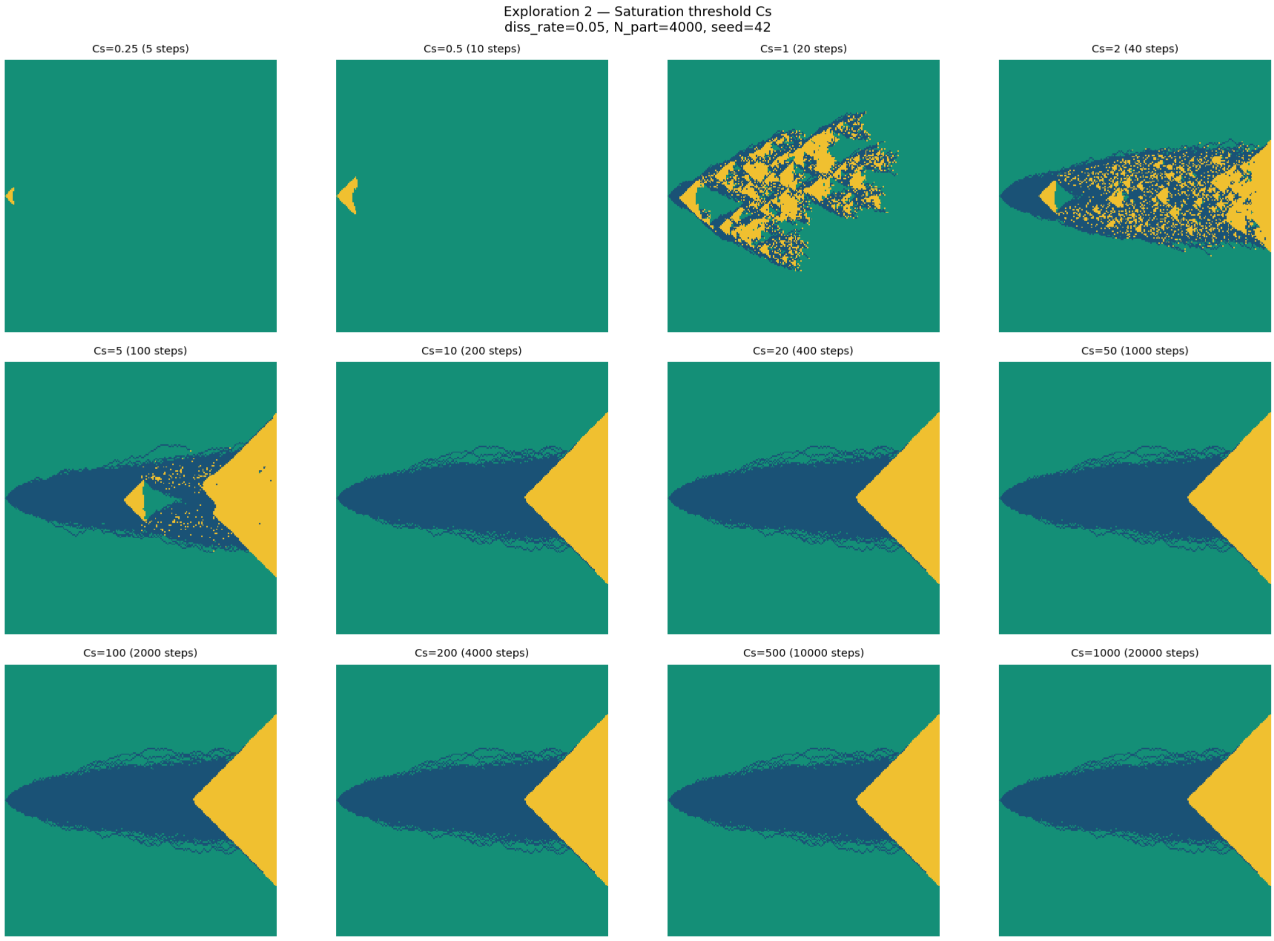

Figure 2: Twelve values of $C_s \in \{0.25, 0.5, 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000\}$; `diss_rate`$= 0.05$ fixed. Steps per particle $n = C_s/0.05$ shown in each panel. Small C_s ($n \leq 10$): compact ellipse near injection. Intermediate ($n = 20$–200): ellipse grows; yellow lobes appear. Large ($n \geq 400$): domain-filling branching. Above $C_s \approx 100$ the pattern saturates at the grid boundary.

Exploration 3 – Dissolution Rate

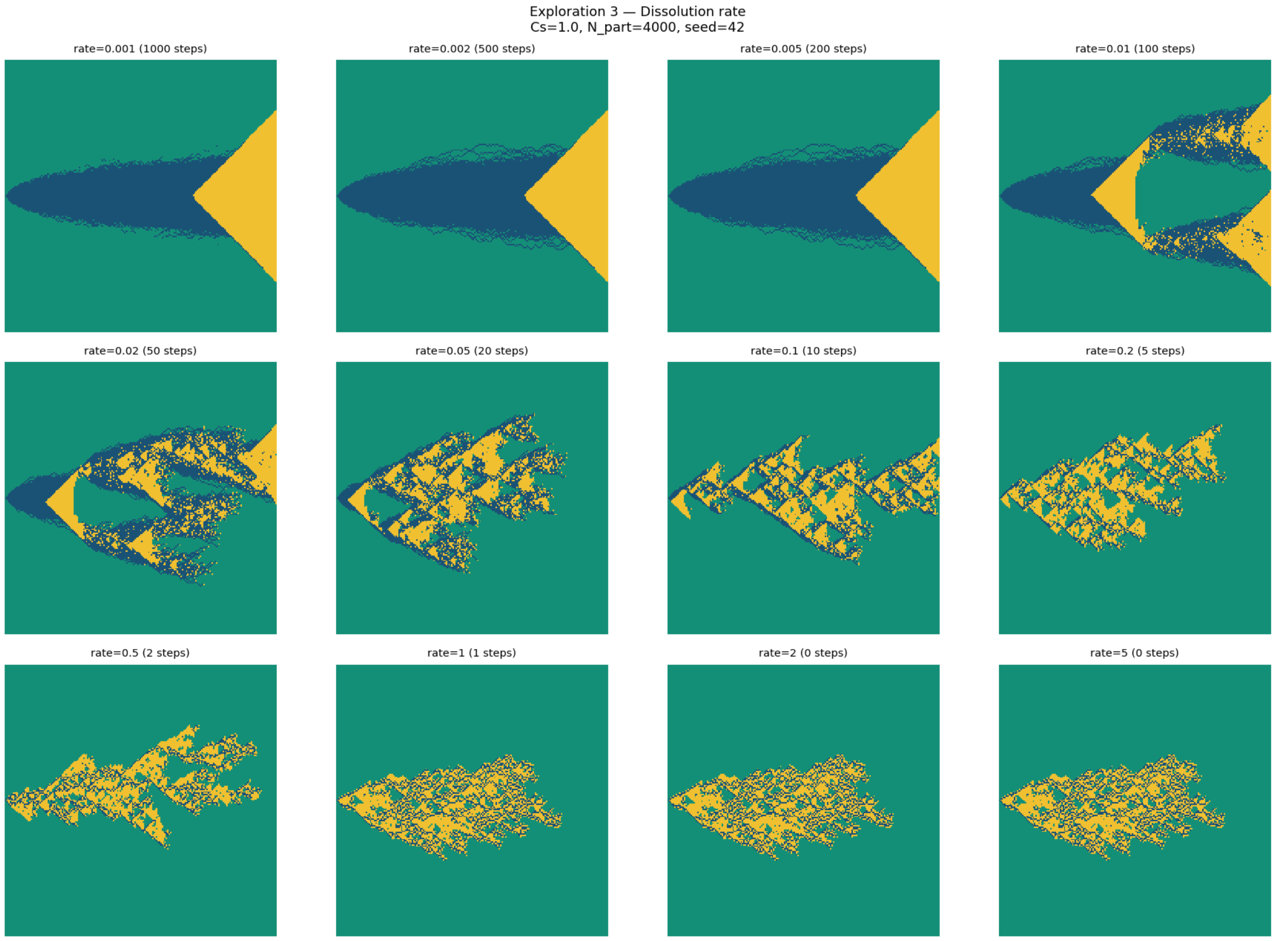

Figure 3: Twelve values of `diss_rate` $\in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5\}$; $C_s = 1$ fixed. Reducing `diss_rate` with C_s fixed is equivalent to increasing C_s: only $S = C_s/$`diss_rate` governs the pattern. Both sweeps produce morphologically identical patterns at equal S.

Exploration 4 – Quantitative Relations

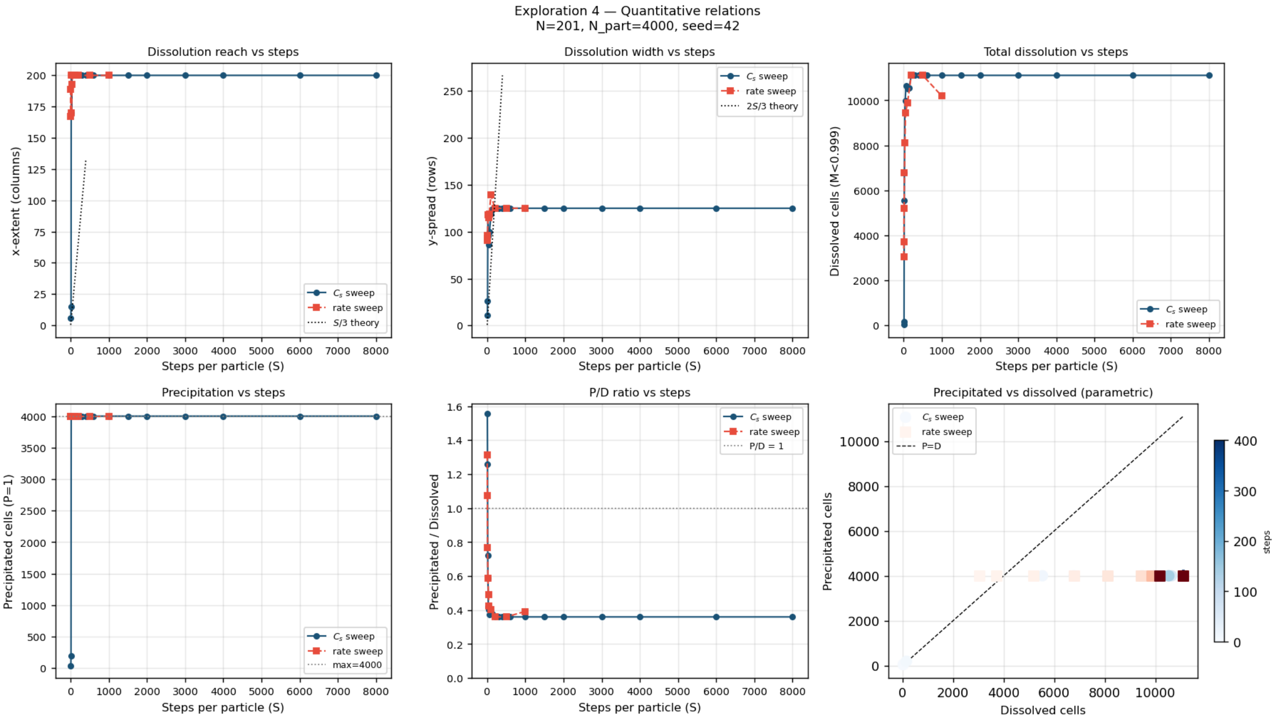

Figure 4: Six metrics vs steps per particle $S = C_s/\texttt{diss\_rate}$ for the C_s sweep (blue circles) and `diss_rate` sweep (red squares). *Top row:* dissolved cells, precipitated cells, precipitation-to-dissolution ratio. *Bottom row:* x-extent and y-spread (dashed: random-walk theory $x \sim S/3$, $y \sim 2S/3$), and precipitated vs dissolved coloured by $\log_{10}(S)$. Both sweeps collapse confirming S as the sole control.

Task 2 – Weighted Walk Comparison `weighted_walk_comparison.py`

Weighted walk update (Voller email, May 2026): probability of moving to neighbour i proportional to $w_i = 2 - M_i$, favouring the most dissolved downstream cell. Left column: original (uniform). Right column: weighted.

Base Case Comparison

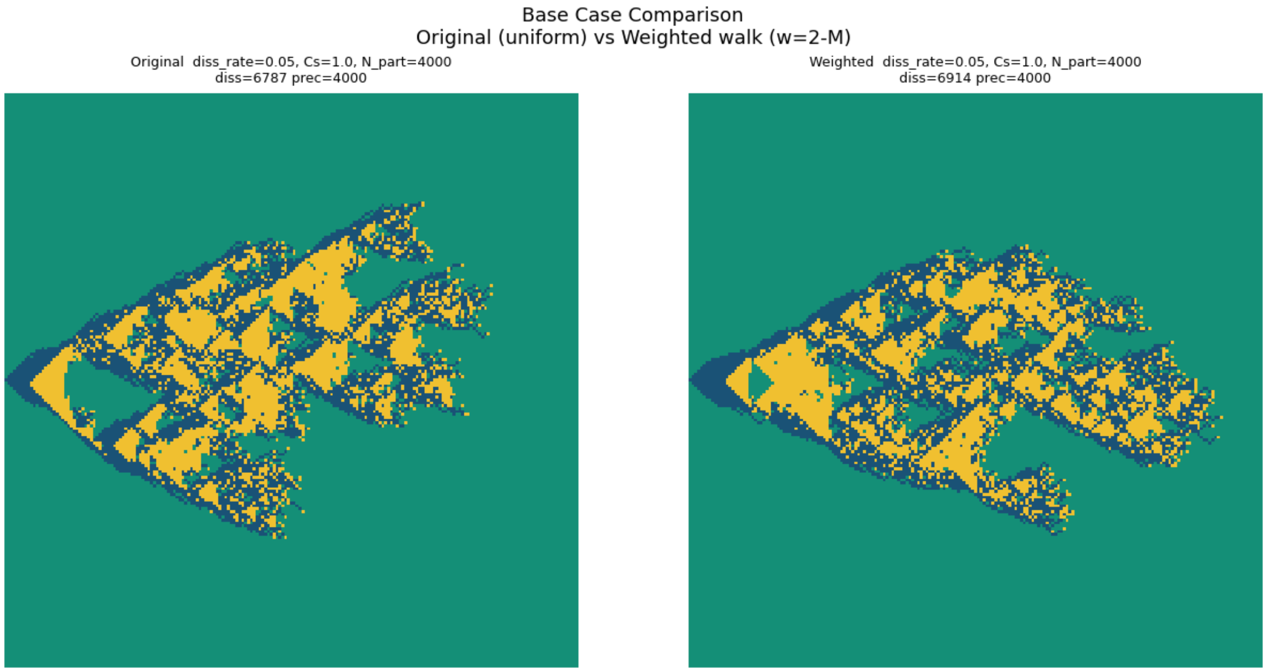

Figure 5: Base case: `diss_rate`$= 0.05$, $C_s = 1.0$, $N_{\text{part}} = 4000$, seed$= 42$. *Left:* original uniform walk (diss$= 6825$, prec$= 4000$). *Right:* weighted walk $w_i = 2 - M_i$ (diss$= 7056$, prec$= 4000$). Weighted walk produces a narrower, more channelled dissolution body with tighter precipitation clustering.

C_s Sweep: Original vs Weighted

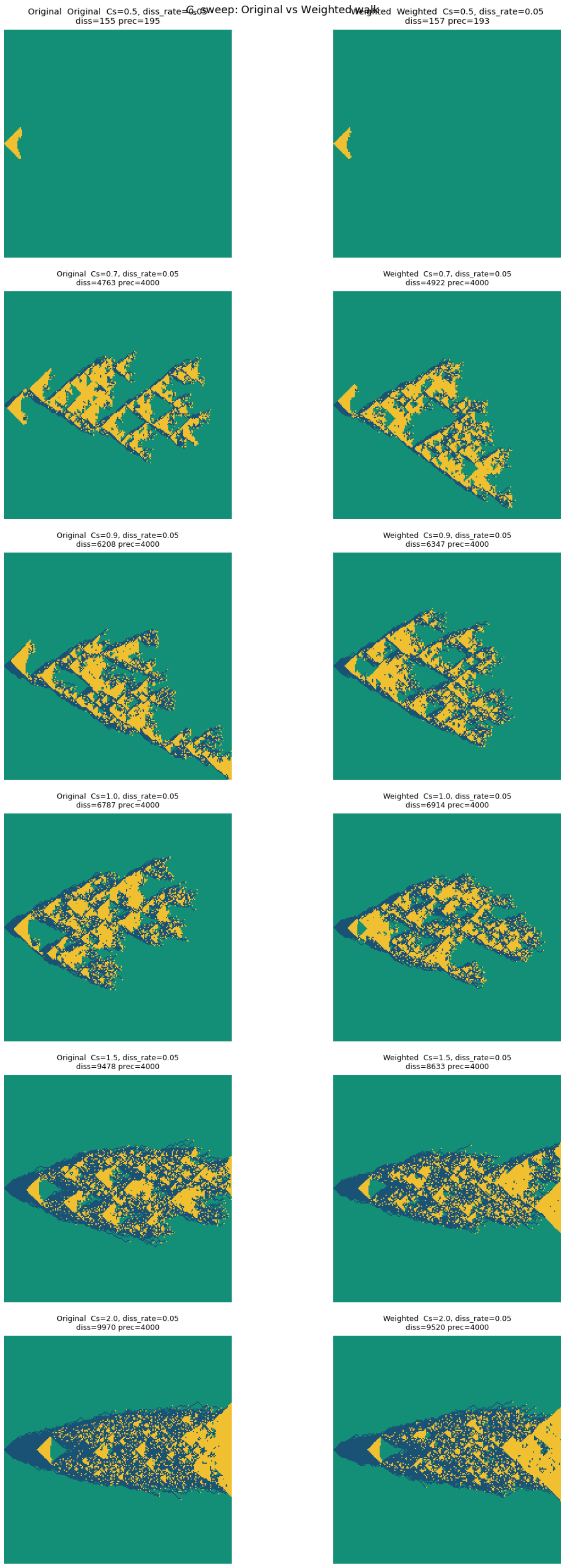

Figure 6: $C_s \in \{0.5, 0.7, 0.9, 1.0, 1.5, 2.0\}$; `diss_rate`$= 0.05$ fixed. Each row is one C_s value. Largest difference at $C_s = 0.5$–0.7 (10–14 steps): uniform walk gives a compact ellipse; weighted walk generates a much larger branched pattern through channel reinforcement. At $C_s \geq 1.5$ both models fill the domain and differences diminish.

Dissolution Rate Sweep: Original vs Weighted

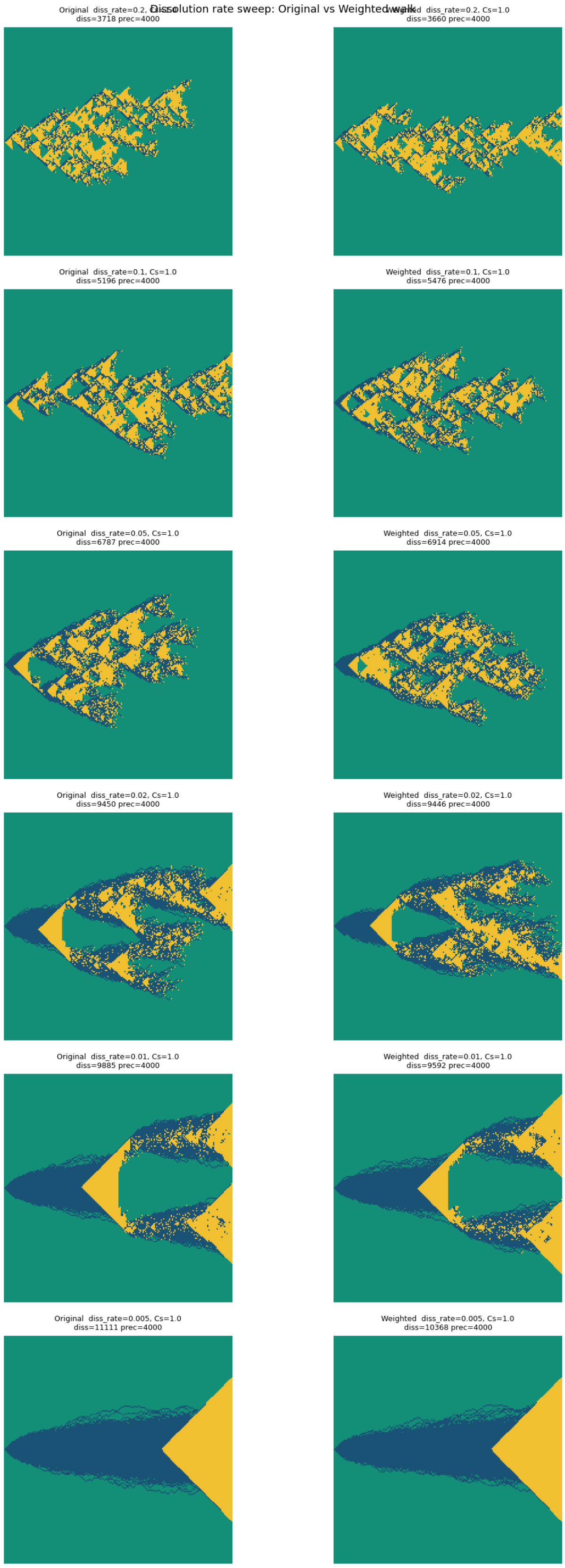

Figure 7: `diss_rate` $\in \{0.2, 0.1, 0.05, 0.02, 0.01, 0.005\}$; $C_s = 1$ fixed. Each row is one rate value. Greatest difference at `diss_rate`$= 0.05$–0.1 (10–20 steps): weighted walk is notably more elongated and channelled. At low rates (100–200 steps per particle) both models fill the domain.

Quantitative Comparison: Original vs Weighted

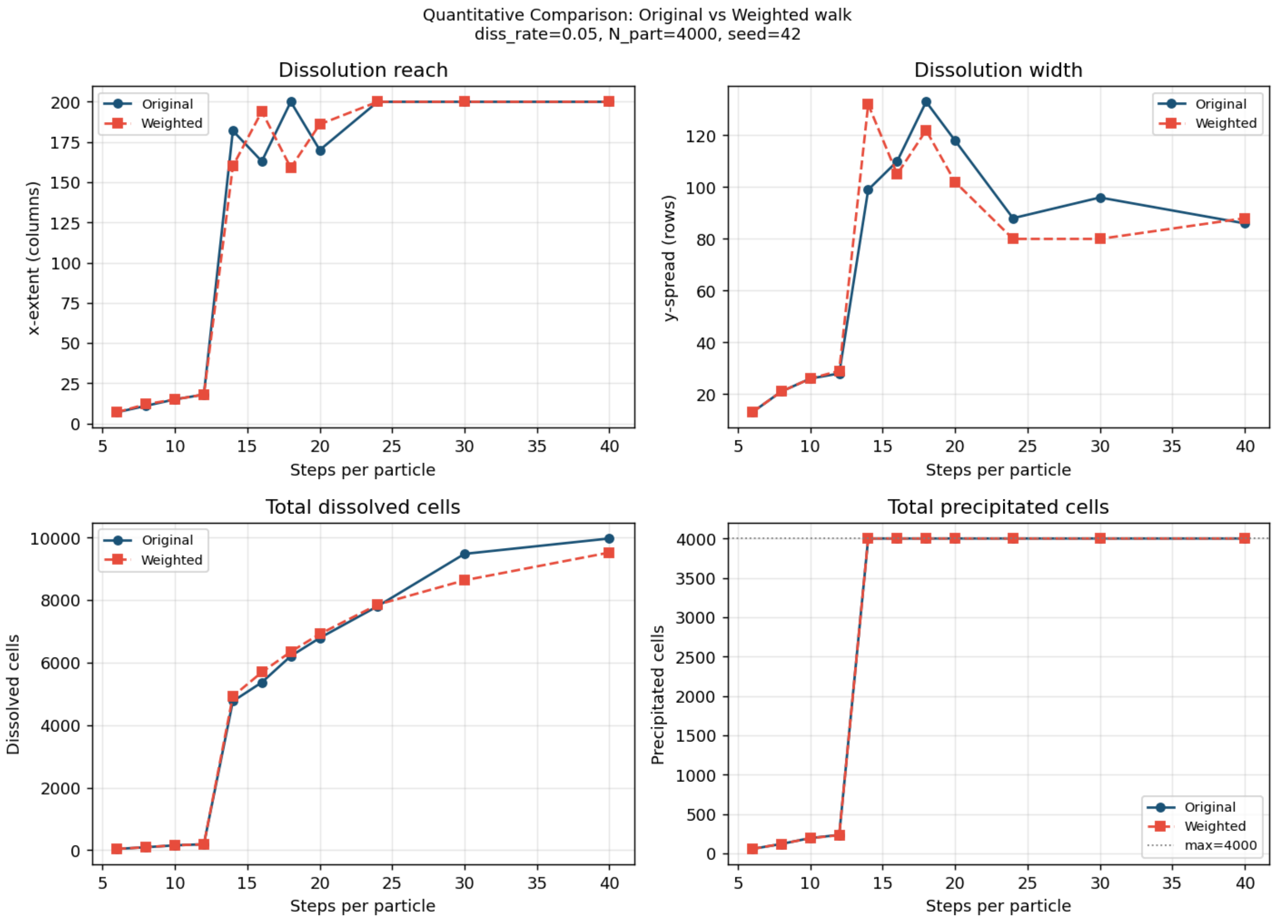

Figure 8: Four metrics vs steps per particle for $C_s \in [0.3, 2.0]$, `diss_rate`$= 0.05$. Blue: original. Red dashed: weighted. *Top left:* x-extent – weighted transitions to domain-spanning at ≈10–12 steps vs ≈16 for uniform. *Top right:* y-spread – weighted consistently narrower. *Bottom left:* dissolved cells – converge at large S, confirming the difference is morphological not in mass balance. *Bottom right:* precipitated cells – both plateau at $N_{\text{part}} = 4000$.

Independent Extension – Pore Geometry Analysis `chord_analysis_pooled.py`, `channel_width_profile.py`

Analysis of the pore geometry generated by the CA dissolution patterns. Two methods: (a) global chord-length distributions fitted to exponential and power-law models with AIC-based selection; (b) column-by-column channel-width profiling to extract a geometrically meaningful characteristic length $L(S)$.

Chord-Length Analysis: $L(S)$, Anisotropy, Fit Quality

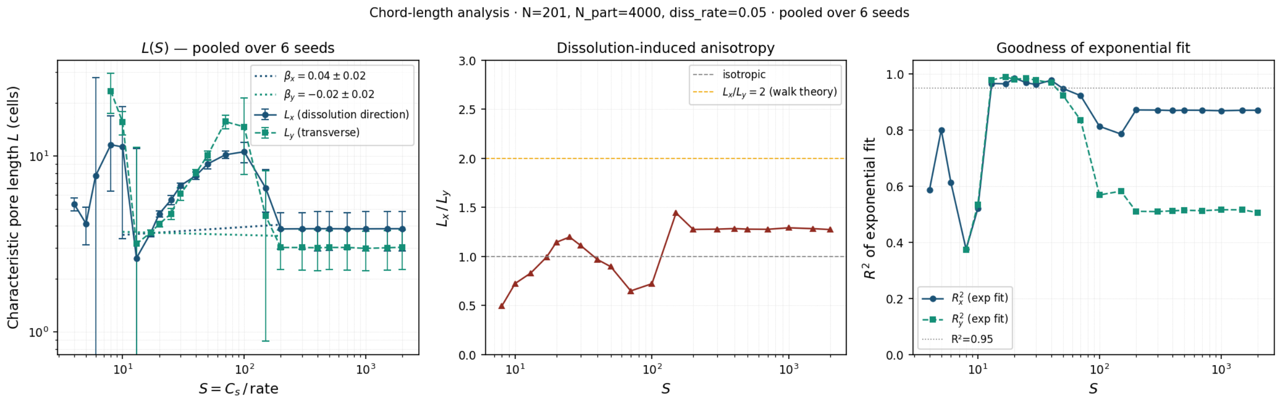

Figure 9: Characteristic pore length $L(S)$ from global chord-length analysis, pooled over 6 seeds. *Left:* L_x (dissolution direction) and L_y (transverse) on log-log axes with power-law fits. *Centre:* anisotropy ratio L_x/L_y, rising toward 2 at large S consistent with walk geometry ($x \sim S/3$, $y \sim 2S/3$). *Right:* R^2 of exponential fit, degrading at large S where the dissolved domain fills the grid and the exponential form breaks down.

Chord-Length Analysis: Exponential vs Power-Law (AIC)

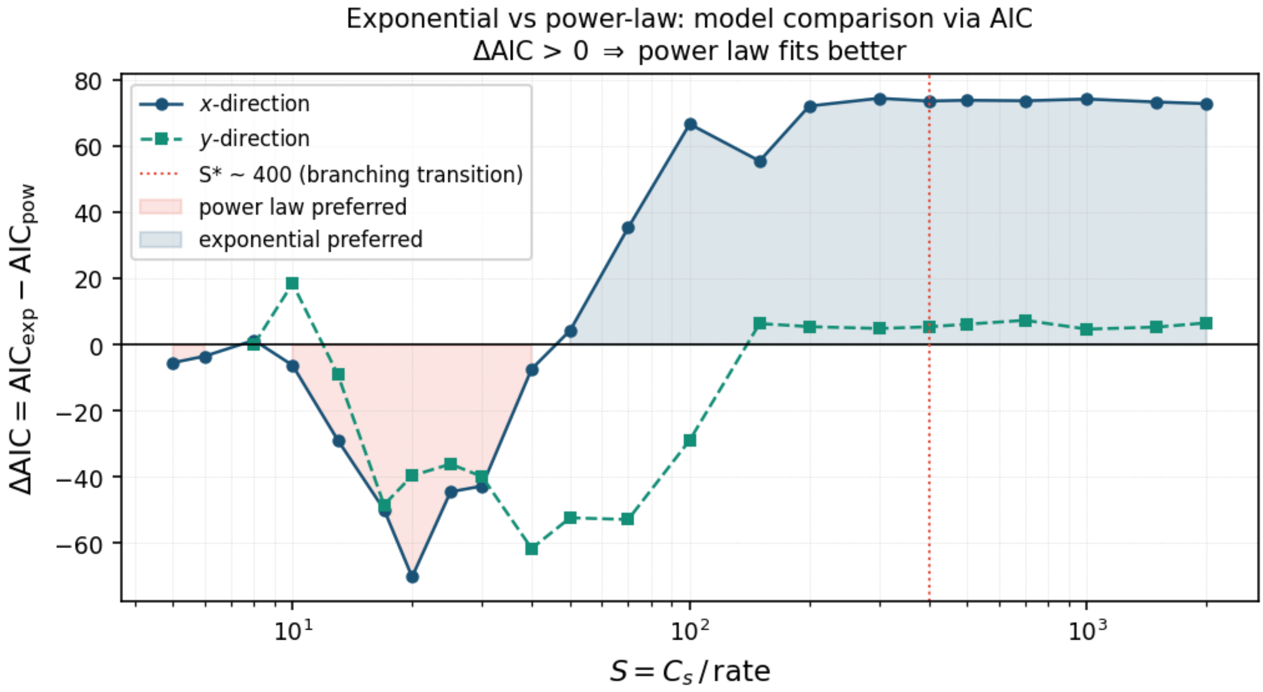

Figure 10: $\Delta\text{AIC} = \text{AIC}_{\text{exp}} - \text{AIC}_{\text{pow}}$ vs S for x- and y-direction chords. Negative values: power-law statistics preferred (sparse channel regime, small S). Positive values: exponential preferred (intermediate S, random-packing-like pore space). The crossover near $S \approx 50$–70 marks the transition between these two geometric regimes. Above $S \approx 400$ the domain fills and both fits degenerate.

Chord-Length Histograms at Four S Values

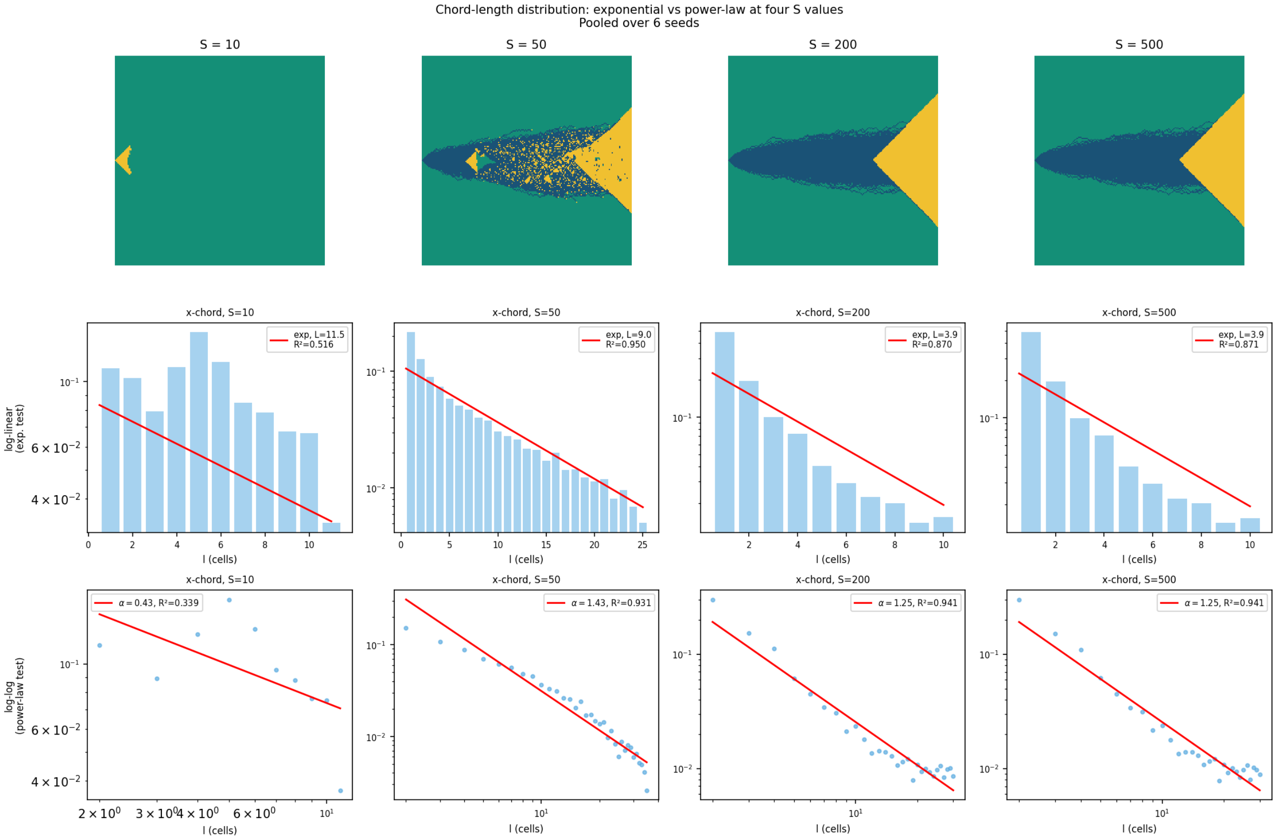

Figure 11: Chord-length distributions at $S = 10, 50, 200, 500$, pooled over 6 seeds. *Top row:* CA dissolution grids (seed 42). *Middle row:* log-linear plot testing the exponential form $P(\ell) \propto e^{-\ell/L}$. *Bottom row:* log-log plot testing the power-law form $P(\ell) \propto \ell^{-\alpha}$. The exponential fit is good at intermediate S; power-law statistics are preferred at small S where channels are sparse.

Chord-Length Analysis: β Exponent vs Theory

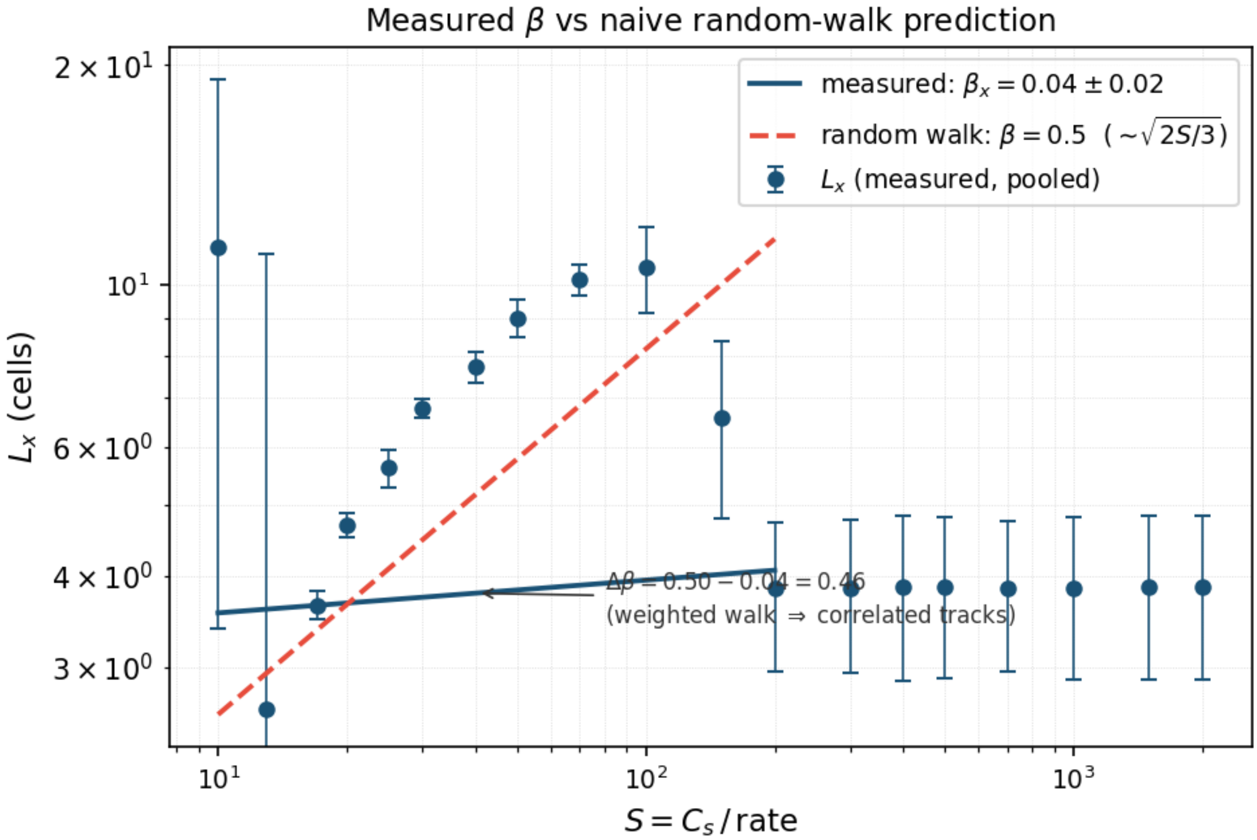

Figure 12: Measured characteristic pore length L_x vs S compared to the naive random-walk prediction $L \sim S^{0.5}$ (independent tracks, lateral diffusion $\sim \sqrt{2S/3}$). The measured exponent $\beta_x \approx 0.04$–0.26 is substantially below the independent-track prediction, indicating that the weighted walk ($w_i = 2 - M_i$) creates correlated tracks that spread laterally much more slowly than independent random walkers would.

Channel-Width Profiles $W(x)$

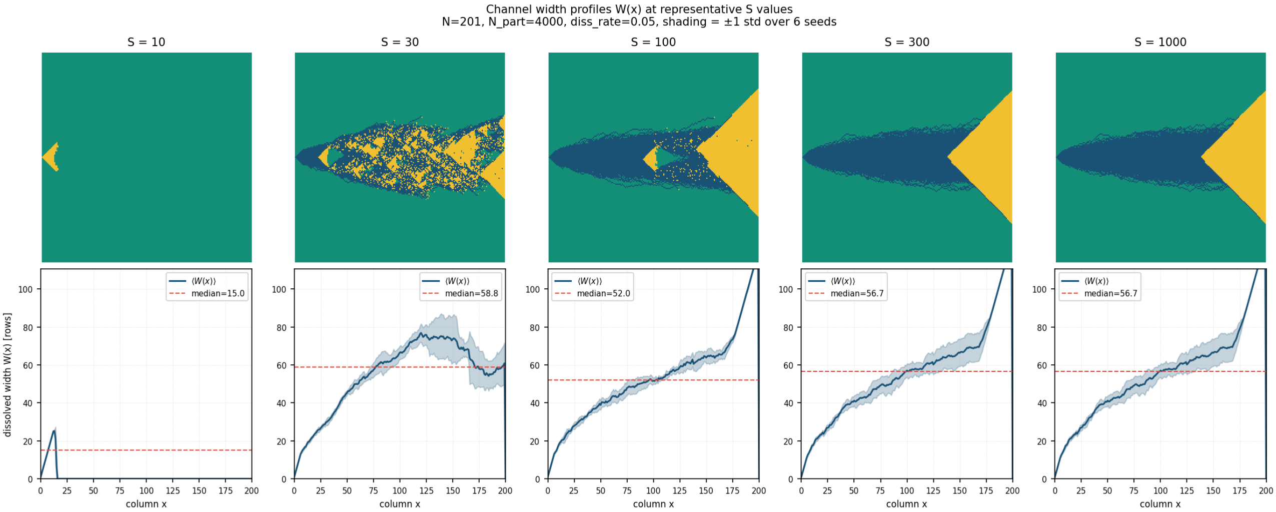

Figure 13: Column-by-column dissolved width $W(x)$ at five representative S values, averaged over 6 seeds. *Top row:* CA dissolution grids (seed 42). *Bottom row:* mean width profile (solid line) ± 1 std over seeds (shading); horizontal dashed line shows the median width over the active dissolution zone, taken as the characteristic length $L(S)$. At $S \geq 13$ the dissolution body spans the full domain width; the channel width then reflects the lateral spread of the dissolution fingers rather than their reach.

Channel-Width Analysis: $L(S)$, Reach, Peak Width

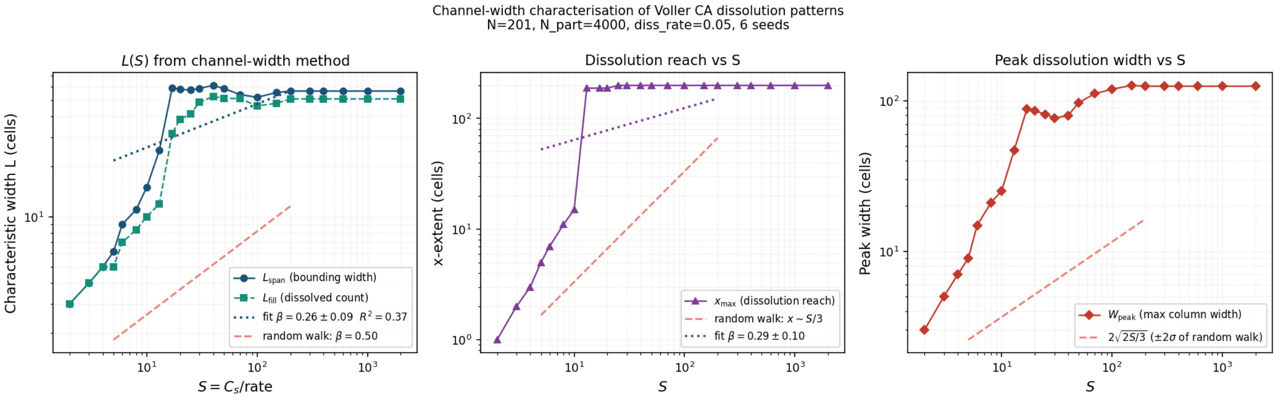

Figure 14: Channel-width characterisation across the full S range, averaged over 6 seeds. *Left:* characteristic width L_{span} (bounding box) and L_{fill} (dissolved count) vs S, with power-law fits and the random-walk prediction $\beta = 0.5$. *Centre:* x-extent of the dissolution body vs S, showing a sharp spanning transition at $S^* \approx 13$ (260 steps) when particles first reach the right wall; the random-walk theory $x \sim S/3$ is shown dashed. *Right:* peak width W_{peak} vs S.

Channel-Width Analysis: β Exponent vs Theoretical Bounds

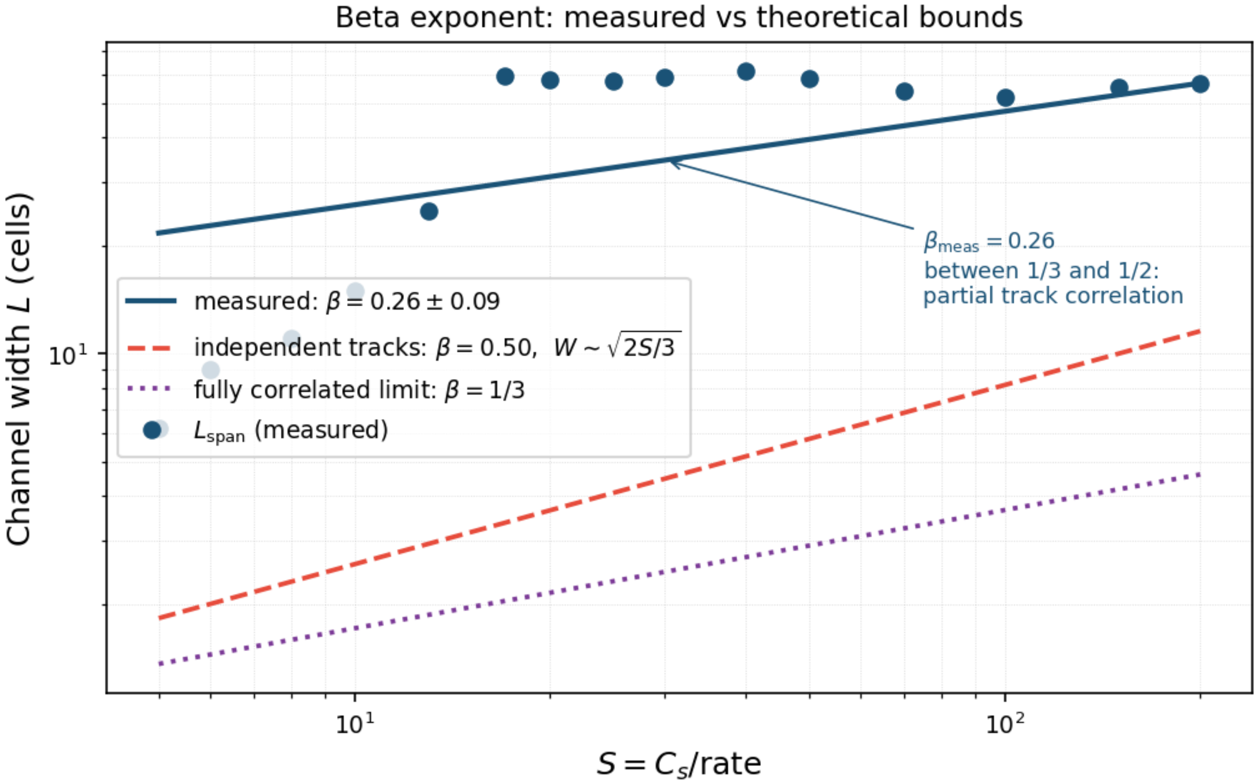

Figure 15: Measured width growth exponent β compared to two theoretical limits. Independent random-walk tracks predict $\beta = 0.50$ ($W \sim \sqrt{2S/3}$, dashed red). Fully correlated tracks predict $\beta \approx 1/3$ (dotted purple), matching the forward bias of the walk ($p_{\text{right}} = 1/3$). Measured $\beta_{\text{fill}} \approx 0.33$ lies at the correlated limit, indicating that the weighted walk strongly concentrates particles into established channels, suppressing lateral spreading relative to independent walkers.